

Experiment No 4 Distillation

- Types of distillation :

1. Simple distillation.
2. Fractional distillation.
3. Steam distillation.
4. under pressure distillation.

- Apparatus :

1. Round flask.
2. Stand 1
3. Thermometer.
4. Rubber connection.
5. Condenser.
6. Stand 2
7. Adapter.
8. Conical flask.
9. Stand 3
10. gas flame.

- Chemicals :

1. water $H_2O_{(l)}$
2. $NaCl_{(s)}$

- Procedure :

1. balance before experiment.
2. the flask "round flask" is clean and take its weight.
3. take 200 ml of water " H_2O " and 1g of " $NaCl$ ".
4. Put the flask on the gas flame or on the stand.

- 5- Connect stand 1 with round flask, and connect the thermometer with the flask by the rubber, connect the round flask with the condenser by the rubber, connect the condenser with the adapter by the rubber, connect the condenser with stand 2
- 6- close the flask
- 7- Start heating up to 100°C
- 8- during heating the gas flame to get the pure water.
- 9- balance after the experiment the flask with the water and take its weight.

- results

1. weight of the water before heating w_1 ?
2. weight of the flask before heating with the water w_2 ?
3. weight of the water after heating w_3 ?

- Aim of experiment
To get pure water.